

Uncertainty in ML Assignment 3 Proposal

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Proposal

In this proposal, we aim to determine whether Monte Carlo (MC) Dropout improves uncertainty estimation and calibration in CNN-based image classification and whether its uncertainty estimates are useful for near out-of-distribution (near-OOD) detection on a larger dataset.

We propose to train a convolutional neural network (CNN) for image classification on the CIFAR-10 dataset. We investigate whether MC dropout during test time improves predictive uncertainty and calibration, and thereafter whether the resulting uncertainty scores can be used to distinguish in-distribution samples from CIFAR-10 images and OOD samples from the CIFAR-100 images.

1 Method

We will develop a CNN in Keras with the Categorical Cross-Entropy loss. We will include dropout layers during training as regularization. At test time we apply Monte Carlo (MC) dropout by keeping dropout active and performing multiple stochastic forward passes on each test image. The mean prediction will be used for classification, while the variance (or entropy) will be used as the uncertainty measure.

To determine whether MC Dropout works effectively, we will compare the MC Dropout CNN model against a baseline standard CNN. We compare the stochastic MC dropout prediction against a deterministic softmax prediction to determine whether MC dropout improves calibration.

Calibration for the models will be evaluated using the Expected Calibration Error (ECE) and compared between the models. We expect the MC dropout CNN to have a lower score than the baseline CNN. In other words, we expect the MC dropout CNN to be better calibrated than the baseline model.

Predictive quality of the models will be evaluated with the Brier score. This will determine the difference in quality of the predicted distributions and whether MC dropout produces a more reliable probability estimate.

2 Metrics

We evaluate classification performance with accuracy and Brier scores. Calibration is evaluated using Expected Calibration Error and visualized with reliability plots. The reliability plots will only be used for in-distribution calibration and will hence only be shown for the CIFAR-10 test case.

For near-OOD detection, predictive entropy and predictive variance will be used as uncertainty scores and evaluated using AUROC, AUPR, and FPR@95%.

3 Datasets

The CIFAR-10 [1] and CIFAR-100 [2] datasets will be used in this study.

CIFAR-10 dataset consists of 10 mutually exclusive classes each with 6000 images. The CIFAR-100 dataset consists of 100 classes and 600 images per class. [2, The CIFAR-100 classes are mutually exclusive with the CIFAR-10 classes].

The MC dropout model will be trained on a subset of the CIFAR-10 dataset and evaluated and tested on separate held-out subsets. Classification and calibration will be done on the CIFAR-10 dataset as well. Near out-of-distribution detection will be evaluated on the CIFAR-10 test set as in-distribution and the CIFAR-100 test set as out-of-distribution.

References

- [1] Will Cukierski. *CIFAR-10 - Object Recognition in Images*. <https://kaggle.com/competitions/cifar-10>. Kaggle. 2013.
- [2] Alex Krizhevsky. *Object Classification Experiments*. Chapter 3 of technical report: Learning Multiple Layers of Features from Tiny Images. University of Toronto, Apr. 2009, pp. 32–35. URL: <https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf>.